ANAS MTAWEH

AI Software Engineer | AI-First Product Builder

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PROFESSIONAL SUMMARY

Computer science graduate specializing in AI-driven product development, with hands-on experience building end-to-end systems using AI-assisted workflows. Experienced in rapidly prototyping, iterating, and refining solutions across backend systems, LLM-based applications, and reinforcement learning environments. Strong in problem decomposition, debugging AI-generated outputs, and translating ambiguous requirements into working software.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, SQL

Backend & APIs: Node.js, Express, REST APIs

AI Systems: LLM Integration, Prompt Engineering, AI-Assisted Development, Reinforcement Learning (PyTorch)

AI Tooling: Gemini, OpenAI API, AI Studio, Ollama, LiteLLM

Data & Storage: MongoDB, SQLite, ChromaDB

DevOps & Cloud: Docker, AWS EC2, AWS S3, AWS ECR, GitHub Actions

Tools & Systems: Git, Streamlit, Flask, Ray, Tree-sitter, NetworkX

PROJECTS

Lokr – AI-Powered Code Intelligence Engine (Graph-RAG)

Technologies: Python, Ollama, Tree-sitter, NetworkX, ChromaDB, Streamlit, LiteLLM, SQLite, Parquet, Git

- Designed and implemented a privacy-first Graph-RAG system enabling local LLMs to retrieve precise, context-aware insights from large codebases.
- Decomposed code understanding into retrieval, parsing, and dependency mapping layers, iterating on multiple architectures using AI-assisted workflows.
- Integrated LiteLLM and an MCP server to dynamically route between local Ollama models and cloud APIs, optimizing privacy and performance tradeoffs.
- Developed a Git-aware incremental indexing pipeline to process only modified files, reducing redundant computation and improving iteration speed.
- Rapidly prototyped and refined a Streamlit chat interface and CLI system by evaluating and improving AI-generated implementations.

Checkers RL – Reinforcement Learning System (D3QN vs AlphaZero)

Technologies: Python, PyTorch, Flask, Ray, CUDA

- Built a reinforcement learning platform for Checkers with an interactive Flask-based web arena and automated tournament evaluation scripts.
- Designed and compared a value-based D3QN agent against a search-based AlphaZero-inspired model using Monte Carlo Tree Search.

- Diagnosed reward hacking and unstable Q-values by iterating on reward design and training dynamics to stabilize learning behavior.
- Implemented distributed training with Ray to enable parallel simulation and faster iteration cycles.
- Evaluated model performance through automated self-play and head-to-head benchmarking.

Mishtika – AI-Powered Pet Care Platform

Technologies: Node.js, React, MongoDB, Docker, AWS EC2, AWS S3, AWS ECR, GitHub Actions, OpenAI API, Jest

- Built and deployed a full-stack application for managing pet profiles, scheduling, and AI-powered care recommendations.
- Designed backend services using Node.js and Express, deployed via Docker to AWS EC2 with CI/CD pipelines using GitHub Actions.
- Integrated an LLM-based assistant using the OpenAI API, injecting structured pet data to generate personalized responses.
- Engineered automated scheduling workflows using node-cron and Nodemailer, ensuring reliable event reminders.
- Iterated rapidly on features using AI-assisted development, validating outputs and refining system behavior through testing.

EXPERIENCE

Production Operator

Jun 2023 – Aug 2023

Liban Jus, Beirut

- Maintained production flow and resolved operational issues in a fast-paced environment.

EDUCATION

Bachelor of Science in Computer Science

Lebanese International University (LIU), Beirut, Lebanon

Software Development, Artificial Intelligence, Database Systems

CERTIFICATIONS

- AWS Certified Cloud Practitioner (2025–2028)
- Introduction to DevOps – IBM & Coursera